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A MODERN PRIMER IN
**PARTICLE &
NUCLEAR PHYSICS**

FRANCESCO TERRANOVA

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Francesco Terranova

University of Milano-Bicocca and INFN

OXFORD
UNIVERSITY PRESS

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Great Clarendon Street, Oxford, OX2 6DP,
United Kingdom

Oxford University Press is a department of the University of Oxford.
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Published in the United States of America by Oxford University Press
198 Madison Avenue, New York, NY 10016, United States of America

British Library Cataloguing in Publication Data
Data available

Library of Congress Control Number: 2021936304

ISBN 978-0-19-284524-5 (hbk.)

ISBN 978-0-19-284525-2 (pbk.)

DOI: 10.1093/oso/9780192845245.001.0001

Printed and bound by
CPI Group (UK) Ltd, Croydon, CR0 4YY

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A Roberta

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Preface

In most universities, students are exposed to an introductory course on nuclear and particle physics at the undergraduate level during the second or third year. It comes as no surprise that this course represents a challenge for most teachers. Modern particle physics is a discipline grounded on quantum field theory (QFT): a tool that is not available to students at the sophomore or junior level. Even when we explain the Standard Model of particle physics focusing on the “discoveries of new particles,” we struggle against the complexity of the experimental techniques devised in the last few decades. As a consequence, we often resort to teaching the fundamentals of particle and nuclear physics building on the historical development of the field: from non-relativistic nuclear systems and the discovery of the first elementary particles up to the Standard Model. Although this history-based approach has stood the test of time, it is a source of confusion for a student joining the field in 2021. Students at the undergraduate level, for instance, are unable to appreciate the straightforward link between nuclear physics and non-perturbative quantum chromodynamics (QCD) because QCD is mentioned at the very end of the course, while nuclear physics and non-relativistic quantum mechanics is presented as the foundation of our understanding of the microscopic world. In a sense, this situation resembles the paradox of many textbooks on classical electromagnetism, where the Maxwell equations are presented without reference to special relativity up to the last chapter of the book.

For particle physics textbooks, there is an additional reason to move away from the historical development of the field. Our students do not discover what a quark is or what the Large Hadron Collider (LHC) is doing by our courses: they hear about them in high school or, more frequently, on the Internet. They attend our courses because they want to *understand* what a quark is or why the LHC has been built. Possibly with the tools they already have: special relativity (SR) and non-relativistic quantum mechanics (QM). In 2016 – the first year I gave this course – I tried to match these expectations and put aside most of the standard textbooks, where either the subject was presented following the historical development or the author sketched the basics of QFT at the beginning of the course. It took a few years to cook up lectures that devise a primer in particle and nuclear physics employing just special relativity and quantum mechanics. Still, most of the students loved this experiment since its inception and helped me to fine-tune the subjects, sharpen arguments and proofs, and prune what was not necessary (or not cute

enough to be presented to undergrads). From 2016 to 2020 my folder has grown from a collection of notes to a real textbook, which is the outcome of the teacher’s efforts plus the ideas (and patience) of many students and colleagues.

On my side, I was delighted to discover how much particle physics can be introduced using, for instance, relativistic kinematics or symmetries in QM, picking up from excellent textbooks or constructing new demonstrations. I had to negotiate between intuitive explanations and rigour, and sometimes I had to give up saying “No way. I cannot explain this stuff using elementary physics.” The narrative of this book has two aims: to provide a solid ground for students willing to join our field at the graduate level and give a modern view of particle physics to students who will not continue to study it any further. The two goals are often in competition and negotiations were harder than expected. The result matches the expectations of a broad readership: undergraduate students in physics in all countries where this course is mandatory, second or third-year physics students where the undergraduate course is optional but chosen by a large audience as an introduction to fundamental physics, Masters students in nuclear or energy and power engineering and, of course, physics enthusiasts at any level.

An overview of the book

We open the course asking “what is an elementary particle” and we set up the scene in Chapter 1, introducing the Standard Model particles and the fundamental interactions – that is, the core of our present knowledge of the microscopic world. We also explain natural units, which are a great tool in particle physics. In general, we urge the readers to go through the examples and the exercises. Some of them are tame, others (e.g. the Planck scale in natural units or – in Chapter 2 – the explanation of why “light is so special in physics”) will give them food for thought. The second chapter is devoted to the language of particle physics: the covariant formalism and the way we describe particle decays and production. This is done gently, keeping the emphasis on similarities with atomic physics without resorting to creation and annihilation operators or other advanced tools. The concepts and, sometimes, the maths of Chapter 2 may be a challenge for students who do not have a background in special relativity: this is why we describe relativistic kinematics and the covariant formalism starting from the basics, relying both on intuitive explanations and a careful derivation of all formulas. The chapter is complemented by Appendix A, where special relativity is briefly summarized. Chapter 3 describes how particles interact with matter and how we observe them. We do not review all particle detectors but we pick a few standard detectors to show what is crucial in the design of particle instrumentation from the basic charge formation mechanisms to the Shockley–Ramo theorem. Once the fundamental concepts of detector physics are covered, we introduce modern techniques in tracking and calorimetry and present the most prominent analysis tools for the

interpretation of particle physics data: the Monte Carlo methods and the corresponding data analysis techniques. We introduce particle accelerators and colliders in Chapter 4. The phase stability theorem – which is the cornerstone of accelerator science – is demonstrated in a very simple way for linear accelerators and the conditions for stability and the limiting parameters are discussed for all types of modern accelerators, with special emphasis on synchrotrons and colliders. This chapter extends the detection techniques of Chapter 3 explaining the methods employed in modern collider experiments. We introduce symmetries in quantum mechanics in Chapter 5. We address this fundamental topic presenting the symmetries as peculiar features of quantum systems that give us information even if we are unable to solve the corresponding equations of motion. This definition emphasizes the prominent role of symmetries in particle physics, where the Hamiltonian is either not known (physics beyond the Standard Model) or leads to equations of motion that are nearly intractable (strong interactions). We then explore the link between symmetries and conservation laws, introduce the P- and C-parities up to admitting that there is no way to define rigorously what the parity of the photon is in non-relativistic QM. We present two classical experiments to determine the parity of the photon and the pion. In this way, we show that we can establish many fundamental properties of the particles even if we do not know yet the underlying theory. C-parity brings us to the world of antiparticles and we introduce, for the first time, relativistic quantum mechanics through a careful analysis of the Dirac equation. During this discussion, we clarify historical interpretations about the nature of antiparticles and introduce the most general and the only consistent explanation of antimatter, which is grounded on QFT and the CPT theorem.

Electromagnetic interactions are discussed in Chapter 6. The chapter is largely aimed at presenting gauge symmetries and explaining why they play such a key role in the description of fundamental interactions. We do not need QFT to display the strength of the gauge principle: we show that even in non-relativistic quantum mechanics, we can build the semi-classical Hamiltonian employing the $U(1)$ gauge symmetry. The main principle of this book is waived in Chapter 6 when we introduce Feynman diagrams as graphical representations of the perturbative series that describe scatterings in quantum electrodynamics (QED). In this case, nothing is proven rigorously but students can appreciate the power of perturbative techniques in one of its simplest and most spectacular applications, the positronium.

Strong interactions are discussed in Chapter 7 as a generalization of QED. We emphasize the difference between QED and QCD using the corresponding Feynman diagrams. We introduce the concept of asymptotic freedom and present the latest findings on confinement. We explain why quarks are real physical particles and how QCD can be experimentally tested, despite the confinement of quarks inside the hadrons. For the first time, here we attack the fundamental problem of the origin of mass, which we will investigate more deeply in later chapters.

We employ symmetries in QM to build the two- and three-quark models in Chapter 8. We stress that even a rough approximation like neglecting the mass of all heavy quarks provides a powerful symmetry for particle classification. This symmetry successfully predicts the C- and P-parities of mesons and baryons, as well. However, we also underline the limitations of the quark models both on mass predictions and the existence of “exotic” bound states, and how these limitations have been superseded in the last ten years.

Nuclear physics is introduced in Chapter 9 from the discovery of the neutron, the isospin symmetry, and the lattice-QCD results on the nuclear potential. In this way, we establish a direct connection between QCD and nuclear reactions. We then analyze the proton–neutron scattering and the classification of nuclear bound states. We do not review all the time-honored nuclear models but we present in detail the shell model that still provides a deep insight into the nuclear structure. In the chapters devoted to nuclear physics, we study not only the science but also the outstanding applications of this discipline. In Chapter 9, in particular, we discuss nuclear reactions as energy generators for (reactors) and against (nuclear weapons) humankind.

Weak interactions are presented in Chapter 10 starting from the conservation of flavor in QED and QCD and its violation in the decay of the lightest particles. The only particle without electric charge and color (the neutrino) is discussed on the empirical ground starting from the three-body decay of the muon. We glimpse the possibility of parity violation using a thought experiment with photons and present the classical experiment of Chien-Shiung Wu quite in detail. The Goldhaber experiment is another great example of ingenuity: it measures the helicity of the neutrino without seeing neutrinos at all. We show in Chapter 10 that the Dirac equation, combined with the findings of Fermi, Wu, and Goldhaber, gives birth to the modern theory of weak interaction, the V-A theory.

In Chapter 11 we discuss nuclear beta decays as an example of weak interactions in bound systems. We extend the discussion to alpha and gamma decays, which are due to an interplay between strong and electromagnetic interactions, and we introduce the radioactive chains. We point out the countless applications of radioactivity and fusion reactions: from radiocarbon dating to the lifecycle of the stars.

The Standard Model of particle physics is introduced in Chapter 12 starting from the contradictions of the V-A theory. We do not attempt any derivation of the electroweak unification but we explain why a unified theory of weak and electromagnetic interactions is unavoidable. Building on the findings of Chapters 2 and 4, we describe the most important discoveries associated with it: the observation of neutral currents in 1973 and massive spin-1 bosons in 1983. Chapter 13 is a bridge to Masters courses. It is devoted to the least known land of the Standard Model: the Higgs and Yukawa sectors, and their observables (CP violation in neutrinos and mesons, particle mixing, and the Higgs boson). These advanced topics are described in some detail using (classical)

relativistic fields and this is the only chapter I would place at the advanced-undergraduate level. Even if it is not an easy reading, all my students loved the chapter because – by theory and experiments – it led them to the frontier of research and shows science in the making. The chapter culminates with the three most important discoveries of the 21st century in our field: the discovery of the Higgs boson, the observation of neutrino oscillations, and the direct detection of gravitational waves. While mentioning these successes, I raise doubts about the beauty and elegance of the Standard Model compared with general relativity and substantiate those doubts pointing an accusing finger to the limitations of both theories. These limitations will be challenges for the next generation of researchers.

Using this book

The prerequisites of this book are a basic knowledge of special relativity and non-relativistic quantum mechanics. As noted above, students lacking this background can fully enjoy the book with little effort: all concepts needed to follow the explanations and the proofs are summarized in Appendix A (Special Relativity), Appendix B (The Principles of Quantum Mechanics) or recalled in Chapter 2. The Further Reading section of Chapter 1 provides suggestions to extend this background (if needed) or deepen knowledge about subjects that are not directly related to particle and nuclear physics.

Each chapter has a set of exercises graded as “easy,” “confidence-building” (one asterisk), and “challenging” (two asterisks). I would like to stress, once more, the importance of these tools since, as with any book that introduces a new discipline, a full understanding of the concepts can be achieved only by practising with concrete problems. In some cases, these exercises are used to introduce new material or fill a gap in proof but this is done only if the exercise contributes to give further insight into the main concepts. “Solutions to selected problems are available in the companion website.” or, if you prefer, Solutions to selected problems are available in the companion website at <https://global.oup.com/booksites/content/9780192845245/>

In my experience, about 70–80% of the material contained in the book can be presented in a one-semester course. The additional material offers a high degree of flexibility to the teacher, who can tune the course according to the audience’s background and teacher’s style and goals. Two common examples are either a course where the students have completed the classes on quantum mechanics or a course where they are taking quantum mechanics concurrently. In the first case, the teacher is free to choose any subject, including the most advanced ones, which are labeled with an asterisk. I suggest selecting topics from Chapters 1–4 and covering more in detail in Chapters 5–12. Note that Chapter 13 is an invitation to Master’s courses and offers a glimpse of the frontier of research in particle physics. In the second case, I recommend placing emphasis on the main concepts of Chapters 1–11 and resorting to the

Appendices every time there is a gap between the lectures and the background of the students. Chapters 12 and 13 can be used as an invitation to more specialized courses at Master's level. In both cases, I encourage the teacher to emphasize Chapters 9 and 11 because they constitute an essential background for undergraduate-level students and give a sense of unity in our discipline (high-energy physics, rare event searches, astroparticle physics, and nuclear physics) that cannot be obtained from any other undergraduate-level course.

I considered this textbook ready for publication when my graduate students headed for these notes every time they got lost in the technicalities of quantum field theory. As with any introductory text, sooner or later, this book will be superseded by your favourite master/graduate textbook. How many times you pick it up from your shelf after the final exam will be the most reliable measurement of its success, and the author is looking forward to your feedback on it.

Acknowledgements

I am indebted to the students that followed my introductory course of particle and nuclear physics in the last five years. The ideas behind this book came from those lectures and I greatly appreciated the comments, suggestions, and enthusiasm I received from them. I am delighted to thank G. Lavizzari, A. Perego, S. Manzini, M. Zanirato, and V. Zito for careful proofreading of the manuscript. I owe special thanks to my colleagues of the Department of Physics “G. Occhialini.” Many of them reviewed parts of the manuscript and gave me very useful suggestions. I hope they will be satisfied with the outcome. I especially thank S. Alioli, M. Calvi, L. Giusti, P. Govoni, C. Oleari, E. Previtali, T. Tabarelli de Fatis, A. Zaffaroni together with N. Charitonidis, C. Giunti, M. Pari, and F. Vissani from INFN and CERN. I wish to express my gratitude to S. Adlung, always unfailingly helpful, and to the OUP team – especially G. Lipparini, F. MacMahon, and J. Walker – that followed the production phase of the manuscript and helped me with figures, language, style, and L^AT_EX issues.

There is no physics book whose acknowledgements do not include thanks to the author’s family for their patience, and this book is no exception. I wish to thank my wife Roberta not only for her patience but especially for the enthusiasm and encouragement to do what I really love – including the writing of this textbook. I thank Alice for her curiosity regarding the content of the book and because, together with her sister, she forbade me from writing more than 450 pages. You will also appreciate Valentina, who drew the funniest figures in this book. My final thank goes to a friend, who reminded me that one does not need to be a Grandmaster to write a good chess manual.

Setting the scene

1

1.1 The dawn of particle physics

We are so used to thinking of matter as made of tiny particles (atoms, electrons, protons, etc.) that we can barely imagine how nature appeared to humans before the inception of the atomic theory. All cultures have been seeking the “fundamental constituents” of matter but, save for a few (and renowned) exceptions, natural philosophers believed that those constituents were continuous media. In ancient Greece, the basic elements were Earth, Air, Water, and Fire. The Vedas added the Ether, while the Chinese replaced Air with Wood. Despite these differences, all things were thought to be the outcome of mixing and interactions of these media. In 1821, A.L. Cauchy provided the first modern definition of continuous medium employing the concept of limit. According to Cauchy, a medium is described by a continuous function – for example, the mass density $\rho(\mathbf{x})$ – that does not abruptly change if you inspect matter with a microscope. Whatever is the magnification power of the microscope, the water looks like water around a space point $\mathbf{x}$ and the density tends to a well-defined value for $\mathbf{x}' \rightarrow \mathbf{x}$. The definition is based on the formal concept of continuity of a function. The famous Cauchy’s “epsilon-delta definition” (Riley *et al.*, 2006) reads:

in a continuous medium, for every $\epsilon > 0$, it exists a $\delta > 0$ and a region of space where the change of density is smaller than ϵ . In particular, for every point $\mathbf{x}'$ located in a neighbourhood of radius δ centered in $\mathbf{x} \implies |\rho(\mathbf{x}') - \rho(\mathbf{x})| < \epsilon$.

The empirical falsification of this definition for *any* continuous medium marks the inception of particle physics. During the 19th century, physicists gathered overwhelming evidence that continuity breaks down at $\delta < 10^{-10}$ m. This tiny distance is the scale of the granular structure of matter: its atomic size.

What is the difference between atomic and particle physics? From the conceptual point of view, there is no difference between the two disciplines. The former embraced the latter at the beginning of the 20th century when Thomson and Rutherford demonstrated that $\delta = 10^{-10}$ m is not the smallest conceivable scale (Buchwald *et al.*, 2013). New granular structures were observed at $\delta = 10^{-15}$ m, which we presently call “nuclei.” The old atomic structures (atoms) were interpreted as bound states of nuclei and electrons. It soon became clear that neither the bound states nor their more fundamental constituents obey the laws of classical physics. In less than 20 years, quantum mechanics and

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relativity were raised to the status of paradigms for the description of the microscopic world.

Modern particle physics is the discipline that studies the fundamental components of matter (particles) and their interactions. To the best of our knowledge, the interactions among particles, the processes of particle creation, annihilation, and decay can be described only by theories that are compliant with special relativity and quantum mechanics. A class of theories that provide full compliance with special relativity (SR) and quantum mechanics (QM) is quantum field theories (QFTs), where particles are described by quantized fields. The Standard Model (SM) is one of the many possible models of the microscopic world built upon the principles of the QFTs. To date, it is the only one that works.

We start our journey in particle and nuclear physics from the most basic question: what is an elementary particle?

1.2 Elementary particles

In classical physics and special relativity, we do not make a distinction between an elementary particle and a point-like particle. The two definitions are equivalent, and macroscopic bodies are assemblies of point-like particles. This is not in contradiction with the existence of the continuous media. To collapse a medium down to a point-like particle, the mass density function $\rho(\mathbf{x})$ must be a function that differs from zero only in one point and is such that:

$$\int_{\mathbb{R}^3} d^3\mathbf{x} \rho(\mathbf{x}) = m. \quad (1.1)$$

The Dirac function does the job:¹

$$\int_{\mathbb{R}^3} d^3\mathbf{x} m \delta^3(\mathbf{x} - \mathbf{x}_0) = m \quad (1.2)$$

where the three-dimensional Dirac delta function $\delta^3(\mathbf{x} - \mathbf{x}_0)$ is defined as:

$$m \delta^3(\mathbf{x} - \mathbf{x}_0) = m \delta(x - x_0)\delta(y - y_0)\delta(z - z_0) \quad (1.3)$$

and describes the mass density of a point-like particle.

Classical point-like particles are by definition elementary particles because they are geometrical points associated with an inertial mass that cannot be destroyed or split.

In non-relativistic quantum mechanics, we can still define a point-like particle but the Heisenberg uncertainty principle introduces a trade-off between particle localization and the determination of the trajectory. If the measurement of the particle position localizes the particle down to a point, we are perfectly measuring the position in space. In this case, the uncertainty on the conjugate variable – the momentum, which includes the direction of motion – is infinite by the Heisenberg principle. We, therefore, know where the particle is but we cannot say anything

¹Well, at the expense of some mathematical subtleties. No ordinary function attains eqn 1.1 if it is zero everywhere but in a point. The Dirac delta function is what we need here and is discussed in Appendix B.

about its trajectory even if we can solve the corresponding equation of motion. We thus need to follow a different path than classical physics.

The motion of a point-like particle in quantum mechanics² is described by the non-relativistic **Schrödinger equation** for a single particle of mass m :

²If you are not acquainted with these quantum mechanics concepts, I suggest reading Appendix B, where you find all the material needed in this book.

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right] \psi(\mathbf{x}, t) \quad (1.4)$$

where $V(\mathbf{x})$ is the potential energy (in the following, potential - see Sec. B.3) describing the forces acting on the particle. The localization is performed with finite precision: the initial wavefunction is a wavepacket that always takes up a non-zero volume. The trajectory is replaced by the time-dependent probability of finding the particle in a finite volume in space. Unlike classical physics, this definition is not equivalent to the definition of elementary particles. The motion of an atom in a gravitational field fulfills this definition but an atom is a composite system and its wavefunction results from the wavefunctions of the constituents. A free **elementary particle** in quantum mechanics should be defined as a system whose wavepacket can *only* be expressed by a superposition of plane waves. But who can ensure a priori that another representation does not exist? For instance, the atom could be a bound state of an electron and a proton (hydrogen) and in this case, the wavefunction at sub-atomic scales is the solution of the Schrödinger equation for the hydrogen. A proton is even more malicious although in most experiments it appears as a harmless point-like particle. The proton is a complex bound system of quarks and gluons (see Chap. 7), whose equations of motion are too complicated to be solved analytically.

The issue does not change in relativistic quantum mechanics and QFTs. The elementary constituents of matter in the Standard Model are the fundamental fields that appear in the Lagrangian of the Standard Model. This definition is as void as the definition based on the wavepacket. We are quite convinced that the Standard Model is part of an unknown general theory and we do not know the fundamental fields of its Lagrangian. To be honest, we do not even know if this theory admits a Lagrangian representation at all. Together with S. Weinberg, we can just say that “the task of physics is not to answer a set of fixed questions about Nature, such as deciding which particles are elementary. We do not know in advance what are the right questions to ask, and we often do not find out until we are close to an answer” (Weinberg, 1997).

All said and done, there is no doubt that we can demonstrate experimentally that a particle is *not* elementary. Experimenters try to decompose candidate elementary particles smashing them or observing properties that demonstrate that the system is composite. To date, we know that all particles are composite except for 25: 6 quarks (and their antiparticles), 6 leptons (and their antiparticles), 12 spin-1 bosons, and 1 spin-0 boson: the Higgs boson.

1.3 Forces, potentials, and fields

Before looking at the particles of the Standard Model, I clear the air on the meaning of “fundamental forces” or, more precisely, **fundamental interactions** among particles.

The paradigm describing interactions among particles has changed (at least) three times from the 17th century to the 20th century and students are exposed to all these paradigms in the course of their academic career. In classical mechanics, the “natural” state of a point-like particle is a uniform motion on a straight line. This motion includes the particle at rest in the laboratory frame as a special case (zero velocity). The origin of a change in the motion of the particle from the natural state is called **force**. The force perturbs the motion under Newton’s second law: $\mathbf{F} = m\mathbf{a}$, where m is the inertial mass of the particle, $\mathbf{a}$ the acceleration and $\mathbf{F}$ the force, a three-component vector applied to the particle. There are many types of forces that can be described by classical physics. Some of them are fundamental forces originating from the intrinsic properties of the particles, others are empirical forces that simplify the description of many-body systems at the macroscopic level.

A renowned example is a gravitational force between two particles that originates by the interplay between the inertial mass of the first particle ($m_1^i = m_1^g$) and the gravitational mass of the second particle (m_2^g). The force on the first particle is:

$$\mathbf{F}_1 = G \frac{m_1^i m_2^g}{r^2} \frac{\mathbf{r}}{r} \quad (1.5)$$

where $\mathbf{r} \equiv \mathbf{r}_2 - \mathbf{r}_1$ is the vector from particle 1 to particle 2 and $r = |\mathbf{r}|$. G is the gravitational constant. This force is attractive, proportional to the product of the masses and inversely proportional to the squared distance between the particles. The direction is along the line that links the two particles: $\mathbf{r}/r$. In this case, $\mathbf{F}_1$ causes the perturbation of the motion of the particle 1 due to the gravitational force generated by the particle 2. General relativity ensures and explains why $m^g = m^i$ for both particles, while this equivalence is accidental in Newton’s theory of gravitation (Cheng, 2010). Using this equivalence, $\mathbf{F}_1$ reads:

$$\mathbf{F}_1 = G \frac{m_1 m_2}{r^2} \frac{\mathbf{r}}{r} . \quad (1.6)$$

The electric Coulomb force generated by a particle 2 on a particle 1 when both are at rest is:

$$\mathbf{F}_1 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \frac{\mathbf{r}}{r} . \quad (1.7)$$

In eqn 1.7, $\mathbf{r}$ is defined as $\mathbf{r}_1 - \mathbf{r}_2$. The Coulomb force has the same functional form as the gravitational law and, therefore, is proportional to the product of the charges and inversely proportional to the squared distance. Unlike gravitation, if the signs of the charges are different, the

force is attractive and if the signs of the charges are the same, the force is repulsive. The issue of the gravitational mass is immaterial, here, because the force is mass-independent. The only mass of eqn 1.7 is the inertial mass of particle 1, which appears in the definition of $\mathbf{F}_1 = m_1 \mathbf{a}$ under Newton's second law. Note that even today, both forces are considered fundamental laws of nature.

On the other hand, the Hooke force is an empirical force that perturbs the motion of a body of mass m through the mechanical contact with a spring (Serway and Jewett, 2018). The force is attractive if the spring is elongated and repulsive if the spring is compressed: $F = -k\Delta x$. k is the stiffness of the spring and Δx is the elongation: if $\Delta x < 0$ the spring is compressed. Hook's force is not a fundamental force of nature and parameterizes the mean effect of the electromagnetic (e.m.) interactions between the body and the spring.

Similarly, the kinetic (or dynamic) friction is defined as the force produced by the friction of two bodies³ – for example, a floor and a moving rigid body with mass m – and is defined as:

$$\mathbf{F} = -\mu_d |\mathbf{N}| . \quad (1.8)$$

In eqn 1.8 and Fig. 1.1, $\mathbf{N}$ is the force perpendicular to the velocity of the body, for example, the weight $|\mathbf{N}| = mg$ of the body. The direction of the frictional force is always opposite to the direction of motion of the body. Again, this is an empirical formula resulting from e.m. interactions and is a typical example of non-conservative forces since the friction reduces the kinetic energy of the body and may bring it to rest.

Newtonian mechanics is suitable to describe a large class of systems, especially at the macroscopic level ($\delta \gg 10^{-10}$ m). Note, however, that the two fundamental forces mentioned above are also the most troublesome for Newton's paradigm: the gravitational law requires an "action-at-a-distance" between bodies that are not in contact. The Coulomb law is in contradiction with Newtonian mechanics when extended to moving charges because the corresponding force – the Lorentz force – is not invariant for Galilean transformations and contradicts Newton's second law (see Chap. 6).

These difficulties can be overcome by the introduction of **fields**: a field is a function that associates a vector to each point in space and may depend on time. If a particle is located in $\mathbf{x}$ at time t , the force perturbing its motion will be uniquely determined by the field at $(\mathbf{x}, t)$ and the intrinsic properties of the particle like the electric charge. Fields were first introduced in electrodynamics by M. Faraday. They are mandatory tools to comply with special relativity, where the speed of any perturbation cannot exceed the speed of light in vacuum. The perturbations are due to the propagation in space and time of the electric ($\mathbf{E}$) and magnetic ($\mathbf{B}$) fields (Griffiths, 2017). In this context, we can define the **potentials** as the minimum number of functions that uniquely define the fields. For the electromagnetic fields, these are the scalar and vector potentials $V(\mathbf{x}, t)$ and $\mathbf{A}(\mathbf{x}, t)$ and:

³Frictional forces tend to stop a moving body and deceived natural philosophers for centuries. Aristotle and most ancient thinkers believed that the natural state of a body without forces was the rest state.

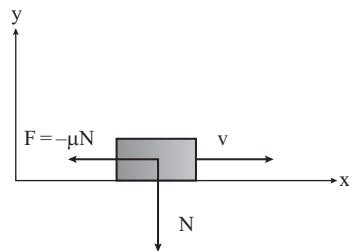


Fig. 1.1 Frictional force $\mathbf{F} = -\mu|\mathbf{N}|$ slowing down a body that moves horizontally on a floor with speed v along the x -axis. $\mathbf{N}$ is the sum of all forces perpendicular to the body velocity. In most cases, $\mathbf{N}$ is just the weight of the body ($|\mathbf{N}| = mg$) and lies along the y -axis.

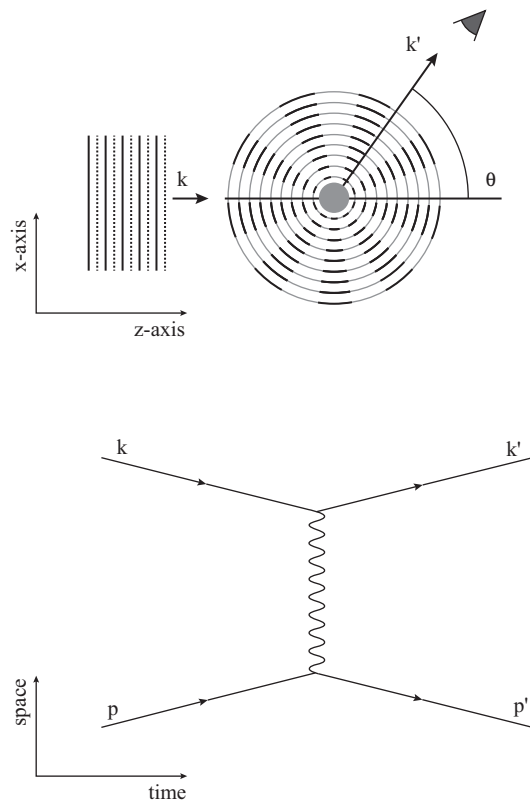


Fig. 1.2 Two-body scattering in QM (top) and QFT (bottom). In the first case, the incoming wavefunction is deformed by the potential of the Schrödinger equation. In the second case, the motion is perturbed by the momentum transferred by the force mediators (see text).

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \quad (1.9)$$

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (1.10)$$

To the best of our knowledge,

all fundamental forces are mediated by fields and are described by a finite number of potentials.

Classical fields and potentials replace the forces in relativistic and non-relativistic QM. The motion of a particle is defined by the Schrödinger equation, which is a function of classical fields and potentials without any reference to forces. The downfall of the concept of forces in physics followed the replacement of classical physics with SR and QM. The new potential-based paradigm describes many particle physics processes employing only the Schrödinger equation or its relativistic extensions. You will love this approach for its simplicity and effectiveness.

The third change of paradigm occurred in the late 1940s when even the concept of wavefunction and potential dropped out of particle physics. The new theory was conceived to account for the creation and destruction of elementary particles: an effect that is ubiquitous in high energy physics. Besides the original motivations, **Quantum Field**

Theories (QFTs) provide a profound understanding of particle interactions at the expense of conceptual (and computational) simplicity. In QFT, wavefunctions and potentials are cast aside by describing particles as **quantized fields** and interactions – sometimes inappropriately called forces – as scatterings among quantum fields. In Fig. 1.2 you can appreciate the difference between the QM description of a scattering and the corresponding description in the QFT framework. Here, the forces are mediated by new particles that transfer momentum to other particles. The reader should be aware that Fig. 1.2-bottom is not only a pictorial representation of the change of the motion of a particle but an actual tool to compute the scattering amplitude in perturbation theory. These kinds of tools are called Feynman diagrams and will be introduced later. In the QFT paradigm:

all fundamental interactions are mediated by quantum fields and the basic observable of any physical process is the transition amplitude between an initial state of N particles and a final state of M particles. In general $N \neq M$ and particles can be created and destroyed during the scattering process.

We will show that these interaction fields are spin-1 particles called **mediators**. Loosely speaking, in QFT the abstract Newtonian forces are replaced by “kicks” given to the particles by the mediators.

1.4 The elementary particles of the Standard Model

Even if nearly all matter in the known universe is composed of protons, neutrons, and electrons, the actual types (in jargon, **flavor**) and the number of elementary particles are significantly larger. Indeed, protons and neutrons do not belong to that list.

The elementary particles of the Standard Model can be classified into four categories: leptons, quarks, spin-1 elementary boson (the force mediators of QFT), and the Higgs boson.

1.4.1 Leptons

Leptons are a set of six spin-1/2 fermions, whose charge and mass are very different among flavors. The first elementary particle discovered belongs to this class and is called the electron. Three leptons are electrically charged. Their charge Q is equal to $-e$, e being the size of the proton charge: 1.6×10^{-19} C. The charged leptons are:

- the **electron** (e^-), a negatively charged fermion with a mass of $511 \text{ keV}/c^2$. As discussed in Sec. 1.5, $1 \text{ eV}/c^2$ corresponds to about $1.78 \times 10^{-36} \text{ kg}$. The electron mass is thus $9.11 \times 10^{-31} \text{ kg}$. The electron is the only stable charged lepton. The vast majority of

electrons in the universe are located in atomic bound states or move freely inside the interior of the stars (plasma electrons).

- the **muon** (μ^-), a negatively charged fermion with a mass of $106 \text{ MeV}/c^2$ or, equivalently, $1.88 \times 10^{-28} \text{ kg}$. It has the same interactions as the electron but a larger mass. The muon is unstable and has a lifetime of $\sim 2.2 \mu\text{s}$ in its rest frame.⁴
- the **tau lepton** (τ^-) is a negatively charged lepton with the same properties of the muon but a much bigger mass ($1777 \text{ MeV}/c^2$) and shorter lifetime: $2.9 \times 10^{-13} \text{ s}$ in its rest frame.

The electrons are among the most abundant particles in the universe because they are stable. Muons are visible even with simple detectors since they are often produced by the interaction of cosmic rays (protons or light nuclei) in the earth's atmosphere and reach the surface of the earth due to time dilation. Tau leptons are much more difficult to be detected and are generally studied using accelerators.

The neutral spin-1/2 leptons are all stable particles and are called neutrinos:

- the **electron neutrino** (ν_e) is a light neutral fermion. It has only two types of interactions: the weak interactions and the gravitational interactions. The mass of the electron neutrino is not defined because the neutrino wavefunction is a linear superposition of three mass eigenstates with mass m_1, m_2 , and m_3 . The exact value of these masses are not known but – as we will see in the forthcoming chapters – are significantly smaller than $1 \text{ eV}/c^2$ (see Fig. 1.3).
- the **muon neutrino** (ν_μ) is a light neutral fermion corresponding to a different linear combination of m_1, m_2, m_3 . It can be distinguished from ν_e because the scatterings that destroy a ν_μ always produce a μ^- in the final state: $\nu_\mu + X \rightarrow \mu^- + Y$, where X and Y are an arbitrary set of particles.
- the **tau neutrino** (ν_τ) is a light neutral fermion corresponding to another linear combination of m_1, m_2, m_3 . The scatterings that destroy a ν_τ always produce a τ^- in the final state.

1.4.2 Quarks

The quarks are spin-1/2 massive fermions with a fractional electric charge. Fractional charges are in clear disagreement with the outcome of Millikan's experiment. They have never been observed in this experiment because they can be found only inside bound states with integer electric charge that are called hadrons. We will provide a profound explanation of this phenomenon in Chap. 7.

- **up quark** (u) is a fermion with a charge equal to $+2/3$ in units of e and a mass of about $2 \text{ MeV}/c^2$ ($3.6 \times 10^{-30} \text{ kg}$). In the universe, it is mostly found in protons, which are composed of two quark up and one quark down (uud) and in neutrons (udd). The only stable particle with an up quark is the proton.

⁴Due to time dilation in special relativity, the lifetime in any other frame is increased by $\gamma = (1 - v^2/c^2)^{-1/2}$. Time dilation is discussed in Sec. A.2.1 and Chap. 2.

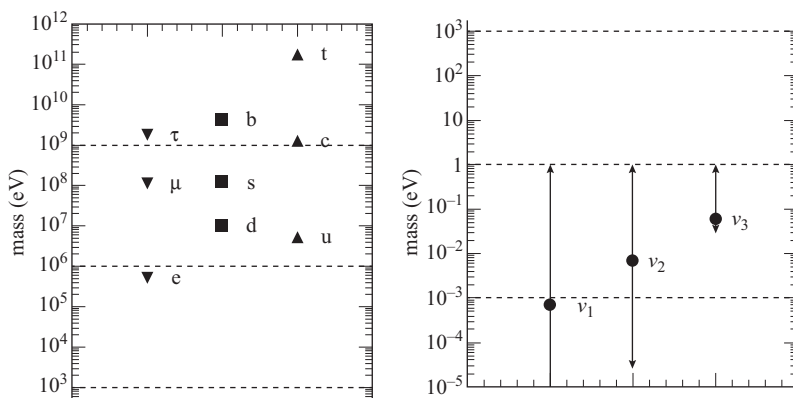


Fig. 1.3 Left: Masses (in eV/c²) of the elementary fermions of the Standard Model. Downward triangles represent charged leptons. Upward triangles show the “up-type” quarks, i.e. the quarks with electric charge equal to +2/3. The squares show the “down-type” quarks with electric charge equal to −1/3. Right: current values and uncertainties of the mass m_1 , m_2 , and m_3 of the neutrinos. Redrawn from (de Gouvea, 2009).

- **down quark** (d) is a fermion with a charge equal to $-1/3$ in units of e and a mass of about $5 \text{ MeV}/c^2$. Again, the only stable particle with a down quark is the proton.
- **strange quark** (s) is a fermion with the same charge as d ($-e/3$) and a mass of about $93 \text{ MeV}/c^2$. No stable bound states contain a strange quark.
- **charm quark** (c) is a fermion with a charge equal to $+2e/3$ and a mass of about $1.3 \text{ GeV}/c^2$. Bound states containing a quark c are always unstable.
- **beauty quark** (b) is a fermion with a charge equal to $-e/3$ and a mass of about $4.2 \text{ GeV}/c^2$ without stable bound states.
- **top quark** (t) is the heaviest known particle ($173 \text{ GeV}/c^2$). It is a fermion with charge equal to $+2/3$ in units of e and decays so quickly ($\tau \simeq 5 \times 10^{-25} \text{ s}$) that cannot form any bound state before decaying.

The masses of the elementary fermions (leptons and quarks) are plotted in Fig. 1.3. The current range allowed for the neutrino mass eigenvalues are labeled ν_1 , ν_2 and ν_3 . The fermions can be divided into $N_f = 3$ **families** made by two leptons of the same type (“electron,” “muon,” or “tau”) and two quarks. The first family is composed by e^- , ν_e , u , d , the second family by μ^- , ν_μ , c , s , and the third one by τ^- , ν_τ , t , b . The Standard Model is mathematically consistent whatever the number of families but it breaks down if a family is not complete. We will explain in Chap. 12 why it is very unlikely that additional families exist. Nonetheless, we do not have explanations for the existence of multiple families ($N_f > 1$) nor any special meaning to ascribe to $N_f = 3$.

1.4.3 Antiparticles

Each elementary fermion has its **antiparticle**: a particle with the same mass, spin, and, if unstable, lifetime but the opposite electric charge. The antiparticles are labeled as e^+ , μ^+ , τ^+ , $\bar{\nu}_e$, $\bar{\nu}_\mu$, $\bar{\nu}_\tau$, $\bar{u}$, $\bar{d}$, $\bar{s}$, $\bar{c}$, $\bar{b}$, $\bar{t}$. We will discuss their origin in Chap. 5. In the Standard Model, the neutrinos

and the antineutrinos are different particles. It is remarkable that after more than 90 years of research we do not have yet an experimental proof that this is actually the case.

1.4.4 Elementary bosons and fundamental interactions

As mentioned in section 1.3, the interactions among fermions are mediated by the exchange of force mediators, which are spin-1 bosons, also called **gauge bosons**. As a consequence, each fundamental force of nature can be classified by its bosons and by the strength of the interaction with the elementary fermions. In the QFT description, the interaction terms are shown in Fig. 1.4 and give rise to:

- The **electromagnetic interactions** mediated by a spin-1 massless particle called the **photon**. This particle is stable and the strength of the interaction is given by the **fine structure constant** α . The photon is also its antiparticle.
- The **strong interactions** mediated by eight spin-1 massless particles called **gluons**. These bosons are stable and the strength of the interaction is given by the fine structure constant of the strong forces: **alpha strong** (α_s). Each gluon coincides with its antiparticle.
- The **weak interactions** are mediated by three spin-1 massive bosons: the W^+ (mass: $80.4 \text{ GeV}/c^2$), its antiparticle W^- and the Z^0 (mass: $91.2 \text{ GeV}/c^2$). The Z^0 is also its antiparticle. The strength of the interaction is different between the $W^\pm$ and Z^0 and also depends on some properties of the scattered fermion. The lifetimes of the weak gauge bosons are extremely short and are of the order of 10^{-25} s .

The strength of an interaction between a boson and a fermion depends on the properties of the fermion. The electromagnetic interactions arise because elementary fermions have an electric charge. Strong interactions originate from strong charges called **colors**, which are owned *only by quarks*. As a consequence,

the most important difference between leptons and quarks is due to strong interactions. Leptons do not interact strongly, while strong interactions are the dominant interaction mode for quarks. The reason for this remains a mystery in particle physics.

⁵The weak charges are called “weak isospin” and “weak hypercharge” and will be discussed in Chap. 12.

All particles undergo weak interactions if they have non-zero “weak charges.”⁵ For neutrinos, weak (and gravitational) interactions are the only possible interaction mode because they are free of electric and color charges. Hence, the strength of the interaction depends on the α ’s, the “charge” of the fermion, and the mass of the gauge boson. In Chap. 12 we will provide a straightforward explanation of why weak interactions are so weak compared with strong and e.m. processes. This is due to the

Question	Refs	
Why does each particle have an antiparticle?	Yes	Chap. 4
Why are weak interactions weak?	Yes	Chap. 12
Why are strong interactions strong?	Yes	Chap. 8
Why are the mass of the fermions so different?	No	Chap. 13
Why are neutrinos so light?	No	Chap. 11
Why can't quarks be seen as free particles?	Yes	Chap. 8
Why is the charge quantized in units of $e/3$?	No	–
Why have the leptons no color charge?	No	–
Why has the SM three families?	No	Chap. 12

Table 1.1 A (quite arbitrary) selection of questions arising by a first inspection of the elementary particles of the Standard Model. The second column (yes/no) indicates if the Standard Model has an explanation for them. The chapter in this book where the topic is discussed is cited in the third column.

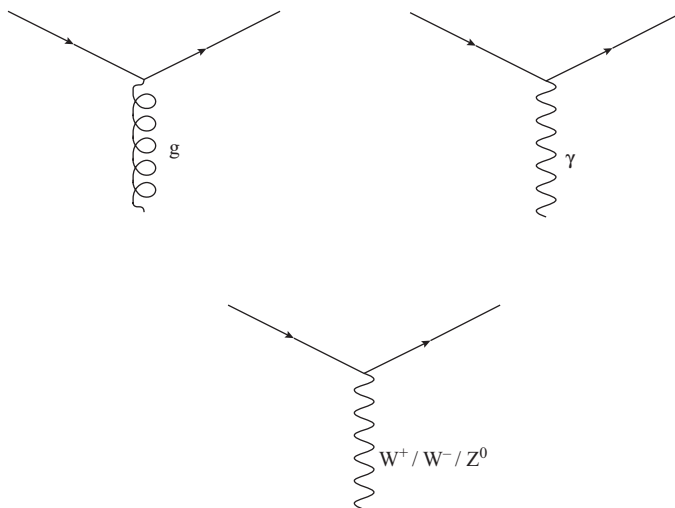
huge mass of the W 's and Z^0 mediators. Fig. 1.4 shows the three types of interactions between fermions and gauge bosons and the approximate values of the α s for a center of mass energy of 300 MeV.

It is interesting to inspect Figs. 1.3 and 1.4 from a fresh perspective: this may be difficult for experienced researchers who have been exposed to these topics for years but it works well for a novice who wants to grasp the strength and limitations of the Standard Model. In Table 1.1, I collected – quite anecdotally – the most frequent questions I was asked after this inspection. At the time of writing, the Standard Model can answer just a fraction of them.

1.4.5 The Higgs boson

The Standard Model also predicts a spin-0 neutral elementary particle: the Higgs boson. Its existence has been disputed for decades because it does not descend straightforwardly from the basic principles of the theory. The **Higgs mechanism** is a technique inherited from other fields (material science and hadron physics) and adapted to particle physics to provide mass to the spin-1 bosons of the weak interactions. The interaction of the Higgs field with the elementary fermions provides masses to these fermions as well. The Higgs is, therefore, a **mass generation mechanism**. Without it, all elementary particles would be very similar to the photon. The mechanism gives firm predictions of the mass of the Z^0 and the W s but lacks predictive power about the mass of the fermions and the Higgs boson itself. The mass generation mechanism for elementary fermions due to the Higgs is called the **Yukawa sector** of the SM and adds a rich set of phenomena that are at the frontier of modern research and will be recapped in the last chapter of this book. The Higgs boson was discovered in 2012 at the LHC collider at CERN

Fig. 1.4 The hierarchy of forces in nature. The figure shows a comparison among the interactions between the fermions (continuous lines) and the mediators (curled lines) that originate the strong (top left), electromagnetic (top right), and weak (bottom) forces. The interactions are mediated by a gluon, a photon, and three massive spin-1 bosons, respectively. The corresponding values of the α s are about $1/137$, 1 , and $1/30$ if measured at a center of mass energy of 300 MeV. The actual strength of weak interactions is further reduced by the mass of the W^+ , W^- , and Z^0 bosons.



Interaction	Boson	Fermions	Strength
Strong	gluons	quarks	1
Electromagnetic	photon	quarks and charged leptons	10^{-3}
Weak	Z^0 , $W^\pm$	quarks and leptons	10^{-8}

Table 1.2 Fundamental interactions of the SM. The corresponding mediators are shown in the second column (“Boson”). Fermions are the types of particles that undergo the interaction. The (relative) strength is an approximate value of the force experienced by two unit-charged particles at a distance of 10^{-15} m normalized to the force due to strong interactions.

and its mass is $125 \text{ GeV}/c^2$. Its lifetime is predicted by the SM to be 1.56×10^{-22} s, although the direct measurements are still limited by experimental uncertainties.

All known fundamental interactions of the Standard Model, its gauge bosons, and the particles that undergo these interactions are summarized in Table 1.2.

1.5 Natural units

Over the years, particle physicists have developed a notation and a system of units that suit their needs and ease compliance with SR. The most widespread system of units in the field is called the **Natural Unit system** and labeled NU.

1.5.1 Metric systems

Physical laws are constraints between physical quantities as length, time, velocity, electric charge, and the like. Only a fraction of these quantities are independent and are used as basic units in metric systems. For instance, the velocity (v) is a ratio between space x and time t and therefore x , t , and v cannot be used simultaneously to define three basic units.

A **metric system** is fixed by the choice of the basic units and the definition of the prototypes that are associated with these units. The first metric system, formalized during the French Revolution, used the meter as the basic unit for space and the prototype was one ten-millionth of the distance from the equator to the North Pole along the earth's circumference.

By far the most important metric system is the International System of Unit (SI), which is enforced by law in nearly all countries (with the notable exception of the United States). The basic units of SI are seven: length, mass, time, electric current, thermodynamic temperature, amount of substance, and luminous intensity. The advantage of modern metric systems is that the prototypes can be reproduced very precisely and do not rely on physical laws. For instance, the modern definition of the meter is based on the fact that it is much easier to measure the velocity of light in vacuum c than the distance from the Equator to the North Pole. Special relativity definitely helps since c is the same for any inertial system, while the Equator to North Pole distance depends on a plethora of subtle geological effects. The current definition of the meter is, therefore, the length of the path traveled by light in vacuum in $1/299792458$ second. In practice, SI employs a velocity as a basic unit to define the length and c is an exact constant (no experimental errors) because is a metric prototype. As expected, time is an independent basic unit in SI. The prototype (1 second – see Fig. 1.5) is defined as the duration of 9192631770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the ^{133}Cs atom. Once more, the frequency of the photon emitted by the corresponding transition in ^{133}Cs is expressed without errors because is a metric prototype. Any other frequency ($\nu = 1/T$) or period (T) will have its own statistical and systematic error. Metric systems are optimized to achieve maximum precision and reproducibility. On the other hand, physicists know that the luminous intensity is not as fundamental as time, and temperature is linked to the mean kinetic energy of a macroscopic body. SI is overly complicated because it ignores well-established physical laws and metric units make the equation of physics more cumbersome than needed.

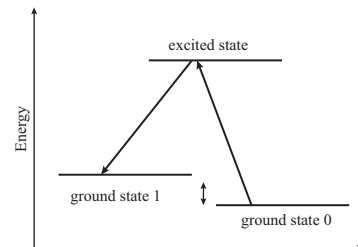


Fig. 1.5 Energy levels of ^{133}Cs employed for the SI definition of time. The hyperfine transition between the lowest states (the double arrow at the bottom of the figure) produces e.m. waves with a frequency of 9192631770 Hz. The “second” is defined by this frequency because of its narrow intrinsic width, which produces a nearly monochromatic line. On the other hand, NU uses a metric prototype only for energy and does not employ a dedicated prototype for time, which results in a loss of precision.

1.5.2 Natural systems of units

Natural units offer an elegant way out. One of the first natural systems was proposed in the 19th century in the framework of Gaussian cgs

(the Gaussian unit system: centimeter–gram–second). It employs the Coulomb law to remove a complicated conversion factor between the electric charge prototype and the other mechanical units. As a consequence, the electric charge is no more a basic unit but a derived unit, and its prototype disappears. The Coulomb law in SI reads:

$$F = k \frac{q_1 q_2}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \quad (1.11)$$

while Gaussian cgs enforces $k = 1$ so that:

$$F = \frac{q_1 q_2}{r^2} . \quad (1.12)$$

Since in the Gaussian cgs, space, time, and mass are measured in cm, s, and g, we have now that one unit charge (“StatCoulomb”) is:

$$1 \text{ statC} = 1 \text{ g}^{1/2} \text{ cm}^{3/2} \text{ s}^{-1} . \quad (1.13)$$

⁶Rationalized or natural systems of units date back to 1881 (G.J. Stoney). The system employed in this book became popular among nuclear physicists during the Manhattan Project (1942–1946). It should be called the “Lorentz–Heaviside natural system of units” but we usually shorten it to “natural units.”

The **natural system**⁶ that is used in particle physics enforces two constraints: the first is inherited from special relativity ($c = 1$) and the second from quantum mechanics ($\hbar = 1$). $c = 1$ removes the unit of space (meter in SI) and its prototype. As a consequence, space could be expressed in seconds because $x = ct$ for a light ray. Similarly, energy could be expressed in kg since $E = mc^2$. However, the basic unit of time is removed by $\hbar = 1$ because $E = h\nu = \hbar\omega$ for a light ray in quantum mechanics (a photon). The prototype of the electric charge can be removed following the procedure of the Gaussian cgs. Particle physicists put $k = 1/4\pi$ (the “Lorentz–Heaviside” choice of k) and the Coulomb law reads:

$$F = \frac{1}{4\pi} \frac{q_1 q_2}{r^2} . \quad (1.14)$$

We are then left with only one unit, which - for practical reasons - is the **electronvolt** (eV). It is the amount of energy gained by the charge of a single electron moving across an electric potential difference of one volt. This unit is commonly used with the metric prefixes milli-, kilo-, mega-, giga-, tera-, peta-, or exa- (meV, keV, MeV, GeV, TeV, PeV, and EeV, respectively). In summary:

the natural system of unit (NU) fixes $c = 1$, $\hbar = 1$, $k = 1/4\pi$, and all physical quantities are expressed in powers of electronvolts (eV).

We warn the reader that there are other natural systems of units currently in use. You may want to get rid of the prototype of the eV setting $G = 1$ in Newton’s law of gravitation. In this case, all units are dimensionless. This system (Planck units), which belongs to the class of “geometrized systems” ($c = G = 1$), is often used in quantum gravity calculations but it is not very common in particle physics and cosmology because G is the most poorly measured fundamental constant.

Natural units simplify enormously the equations of physics but they come with shortfalls we should acknowledge. The most important one is that these units are not well suited for metrology, that is for precision tests of fundamental physics, because they imply the validity of some physics laws. Furthermore, their units are not based on *the best* quantities that can be measured in a lab. What does the “best quantity” in metrology mean? Sometimes, we can measure with outstanding precision observables that are not particularly meaningful for physics like macroscopic currents instead of the electron charge or the hyperfine atomic transitions instead of time differences (see Fig. 1.5). These ultra-precise measurements are not made to enter the Guinness World Records but to extract the best estimate of a fundamental unit indirectly. The SI prototypes are chosen among these special observables. We cannot do the same in NU if we decide to remove all prototypes except energy. To kill two birds with one stone and get the best out of metric and natural units, we perform our calculations in NU and, then, we return to SI using a set of conversion rules.

1.5.3 Conversion rules

Conversions between NU and SI can be done easily employing dimensional analysis. A formula in NU is proportional to the corresponding formula in SI:

$$(NU) = \hbar^a c^b (SI) \quad (1.15)$$

where a and b are rational numbers that come from the dimensional analysis of NU and SI. For instance, the relativistic energy of a particle is $E = \gamma mc^2$ in SI and $E = \gamma m$ in NU (see Chap. 2). Hence, $a = 0$ and $b = -2$. This can be derived from eqn 1.15 since γ is dimensionless and E is expressed in Joule ($\text{kg m}^2/\text{s}^2$) in SI. The dimension of $\hbar$ ($[\hbar]$) is $[M][L]^2[T]^{-1}$ and $[c]$ is $[L][T]^{-1}$. $[M]$, $[L]$ and $[T]$ are the dimensions of mass, length, and time, respectively. Then,

$$\begin{aligned} [M] &= [M][L]^2[T]^{-2}[ML^2T^{-1}]^a[LT^{-1}]^b \\ &= [M]^{1+a}[L]^{2+2a+b}[T]^{-2-a-b}. \end{aligned}$$

This equation is satisfied if $a = 0$ (from $1 = 1 + a$) and $b = -2$ (from $2 + 2a + b = 0$). The third constraint ($-2 - a - b = 0$) is automatically satisfied. The reader should not be surprised to find an over-constrained system of three equations and two unknowns. The third equation signifies that $E = \gamma m$ is dimensionally correct in NU!

To convert numerical values from NU to SI, we just need the conversion factor between Joule and eV:

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}. \quad (1.16)$$

A particle with a 511 keV (kilo-eV) mass (the electron) corresponds to an energy of $511 \times 10^3 \cdot 1.6 \times 10^{-19} = 817 \times 10^{-16} \text{ J}$ and a mass of $817 \times 10^{-16}/(3 \times 10^8)^2 = 9.08 \times 10^{-31} \text{ kg}$. Giving up the approximations and using the world best measurements of the electron

mass, $m_e = 510.9989461(13)$ keV, corresponding to $9.10938356(11) \times 10^{-31}$ kg. In this expression, (13) and (11) indicate the experimental uncertainty in the last two digits: 61 and 56, respectively. Particle physicists use both the parenthesis notation and the $\pm$ notation ($m_e = 510.9989461 \pm 0.0000013$ keV) to specify the uncertainty. The former is particularly neat for high-precision measurements and is widely employed in metrology.

Another drawback of NU is that the system automatically subsumes dimensional analysis and it is impossible to distinguish among, say, energy, momentum, and mass from the dimension of their units. A useful trick that sidesteps this ambiguity is to write down the c factors that appear in SI: the energy is expressed in eV, the mass in eV/c^2 , and the momentum in eV/c . A $1 \text{ GeV}/c$ particle is thus a particle with a *momentum* of 1 GeV. Note that the “ eV/c ” unit is just a symbol. In NU, the momentum is measured in eV like the energy and the mass.

Converting NU and SI values is straightforward in most of the cases and you do not need to go through eqn 1.15 if you remember a couple of conversion factors:

$$\hbar c \simeq 197 \text{ MeV} \cdot \text{fm} \quad (1.17)$$

$$\hbar \simeq 6.58 \times 10^{-16} \text{ eV} \cdot \text{s} \quad (1.18)$$

where fm (femtometers or Fermi) corresponds to 10^{-15} m. Equation 1.17 converts lengths from NU (eV^{-1}) to SI (m) and eqn 1.18 converts times from NU (eV^{-1}) to SI (s). You can find two examples below: the former is trivial and does not require eqn 1.15. The latter is more sophisticated.

Example 1.1

The Compton wavelength of a particle is equal to the wavelength of a photon, whose energy is the same as the mass of the particle. For the electron case,

$$\lambda = \frac{h}{m_e c} = \frac{2\pi\hbar}{m_e c} \quad (1.19)$$

In SI, the wavelength is $6.63 \times 10^{-34} \text{ J} \cdot \text{s} / (9.11 \times 10^{-31} \text{ kg} \cdot 3.00 \times 10^8 \text{ m/s}) = 2.42 \times 10^{-12} \text{ m}$. In NU:

$$\lambda = \frac{2\pi}{m_e} = \frac{2\pi}{511 \text{ keV}} = 12.29 \text{ MeV}^{-1} \quad (1.20)$$

You can convert NU into SI using directly eqn 1.17 and sidestep the full analysis of eqn 1.15.

$$\lambda_{SI} = 197 \text{ MeV} \cdot \text{fm} \cdot 12.29 \text{ MeV}^{-1} = 2421 \text{ fm} = 2.42 \times 10^{-12} \text{ m}. \quad (1.21)$$

The next example is a case where we need to employ eqn 1.15.

Example 1.2

The Planck mass is a cornerstone of quantum gravity. It is defined in NU as:

$$M = \frac{1}{\sqrt{G}} \quad (1.22)$$

with G being the gravitational constant. We know G in SI ($6.67430 \pm 0.00015 \times 10^{-11} \text{ m}^3\text{kg}^{-1} \text{ s}^{-2}$) and we are using a natural system (NU) where $G \neq 1$. What is the value of G in NU and the value of M in SI? In this non-trivial case, we resort to eqn 1.15: $M = \hbar^a c^b G^{-1/2}$. The dimensional analysis provides the values of a and b :

$$[M] = \left[\frac{ML^2}{T} \right]^a \left[\frac{L}{T} \right]^b \left[\frac{L^3}{MT^2} \right]^{-1/2} = [M]^{a+1/2} [L]^{2a+b-3/2} [T]^{-a-b+1} \quad (1.23)$$

which gives

$$\begin{aligned} a + 1/2 &= 1 \\ 2a + b - 3/2 &= 0 \\ 1 - a - b &= 0 \end{aligned}$$

This is an over-constrained system of equation solved by $a = 1/2$, $b = 1/2$. The Planck constant in SI is then:

$$M = \sqrt{\frac{\hbar c}{G}} = \sqrt{\frac{1.05 \times 10^{-34} \text{ J s} \cdot 3.00 \times 10^8 \text{ m/s}}{6.67 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}}} = 2.17 \times 10^{-8} \text{ kg} \quad (1.24)$$

The Planck mass is just a few tens of micrograms but it is enormous if expressed in NU. $2.17 \times 10^{-8} \text{ kg}$ corresponds to an energy of $2.17 \times 10^{-8} \text{ kg} \cdot (3.00 \times 10^8 \text{ m/s})^2 = 1.95 \times 10^9 \text{ J}$. Since 1 eV is $1.6 \times 10^{-19} \text{ J}$, the Planck mass corresponds to $1.22 \times 10^{28} \text{ eV}$, which is about 15 orders of magnitude the energy reached by the most powerful accelerator in the world (the LHC). In turn, eqn 1.24 provides G in NU: $6.7 \times 10^{-57} \text{ eV}^{-2}$.

The SI value of M shows how difficult is to compress the energy of a macroscopic body - in this case, the weight of a few thousand human cells - into the volume filled by an “elementary particle.”

Further reading

This book requires a basic knowledge of special relativity and quantum mechanics but can be enjoyed even without such background with little effort. All SR concepts needed in the forthcoming chapters are recalled in Appendix A

and Chap. 2. If you want to appreciate the elegance of SR, I suggest reading a few introductory chapters among the vast literature on SR. For instance, Chaps 2, 3, 5, and 6 of [Rindler, 2006](#) provide a clear introduction to the subject. Similarly, Chaps 2 and 3 of [Cheng, 2010](#) give a succinct but effective background. A gentler introduction is given by [Wittman, 2018](#). Appendix B provides the basic concepts and tools of quantum mechanics that are used in this book. To delve deeper, there are many outstanding textbooks. A solid background is given in

Chaps 1–7 of [Griffiths, 2018](#), for example. If you already have a good knowledge of QM, you may want to dig into the links between the founding principles of QM (symmetries, measurements, perturbation theory, entanglement, etc.) and the corresponding applications in particle and nuclear physics. You will find inspiration and alternative points of view in [Weinberg, 2012](#); [Griffiths, 2018](#); [Sakurai and Napolitano, 2017](#); [Manousakis, 2015](#); and [Ballentine, 2014](#).

Exercises

(1.1) How many types of interactions contribute to the scattering of two quarks?

(1.2) Suppose that quarks have the same color charge. Is the total interaction probability of a quark u and a quark d the same as for two d quarks?

(1.3) Consider the following processes:

$$\begin{aligned} n &\rightarrow p + e^- + \bar{\nu}_e \\ e^- + \nu_e &\rightarrow e^- + \nu_e \\ e^- + p &\rightarrow e^- + p \\ e^- + e^+ &\rightarrow e^- + e^+ \end{aligned}$$

How many of these processes need a full QFT description?

(1.4) If we assume the kinetic theory of gases, what is the unit of temperature in NU?

(1.5) The proton Compton wavelength is defined as

$$\lambda_p = \frac{h}{m_p c}$$

in SI. Compute its value both in SI and NU.

(1.6) Compute the e.m. force between two protons located a 10^{-10} m and compare this value with the corresponding gravitational force.

(1.7) Demonstrate that λ_p/λ_e in SI has the same numerical value of λ_p/λ_e in NU.

(1.8) What is the unit of the wavefunctions in NU? [Hint: inspect the definition of wavefunction in Appendix B.]

(1.9) Is it possible to extend NU in such a way that the unit of temperature is eV, even beyond the kinetic theory of gases?

(1.10) * What is the value of the electron charge in NU?

(1.11) * Demonstrate that electric and magnetic fields have the same dimensions in NU.

(1.12) ** In an imaginary universe, suppose that the maximum velocity of a body is c , as in SR, but the maximum energy of a photon must be finite according to this hypothetical formula:

$$\begin{aligned} E &= \hbar\omega & \text{if } \hbar\omega \ll \Lambda \\ E &\rightarrow \Lambda & \text{if } \hbar\omega \rightarrow \Lambda \end{aligned}$$

where Λ is an energy. Can you employ NU to describe this universe? Is there a better natural system that can be used to get only dimensionless units?

Scattering and decay

2

2.1 Covariance

Particle physics is rooted in special relativity (SR) and the principles of the theory are embedded in the units ($c = 1$) and language of particle physicists. This choice comes as no surprise: SR requires a generalization (general relativity, GR) only when gravitational interactions play a role in the dynamics of a system. This situation is extremely rare in particle physics, where other interactions are overwhelming and the distances traveled by particles are so small that space-time can always be considered locally flat. In this framework, SR can be easily employed to handle non-inertial systems and accelerating particles too (see Sec. A.7). Exceptions arise, for instance, when considering the motion of a particle at cosmological distances or in the proximity of a black hole, where a description based on GR is mandatory.

The concept of covariance can be introduced on a general ground to cope with SR, GR and quantum gravity and can be restricted to SR in all cases of interest for this book.

Covariance is a prescription on the form of the physical laws that ensures compliance with the axioms of relativity:

(A1) The physical laws retain the same functional form in all inertial frames.

(A2) The speed of light in vacuum is the same for all inertial observers.

Without a covariant formalism,¹ we should check explicitly that all quantities describing a system (constants, vectors, or any other combination) linked by a physical law such as $f(a, b, c, \dots) = 0$ change in the right way after a Lorentz transformation (LT). The “right way” is:

$$f(a, b, c, \dots) = 0 \xrightarrow{LT} f'(a', b', c', \dots) = 0 \quad \text{AND} \quad f' = f \quad (2.1)$$

so that the physical law in the new reference frame reads:

$$f(a', b', c', \dots) = 0 \quad (2.2)$$

where a', b', c' are the transformed quantities.

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¹In some SR textbooks, the term covariant is used more subtly. A “covariant law” is a law consistent with SR. An “explicitly-covariant law” is a law written in such a way that the consistency with SR can be assessed at first sight, i.e. without using the Lorentz transformations. For sake of clarity and generality, we prefer to use “covariant” and “explicitly covariant” as synonyms. This choice implies that the physical laws can be classified into three groups: laws compatible with SR, incompatible with SR, and covariant. We can thus recast in a covariant form any law compatible with SR.

2.1.1 Covariant and contravariant quantities

Modern geometry describes and classifies objects based on how they change under a general class of transformations (rotations, translations, projective transformations, etc.). The quantities that are interesting in physics may change if the reference system is changed by one of these transformations but – to the best of our knowledge – reference frames and coordinates do not exist in nature: they are practical tools used to describe the physical systems. Physical laws should always be independent of the specific choice of the coordinates.

General covariance is the invariance of the form of the physical laws under the action of differentiable (smooth) coordinate transformations.

For concreteness, consider a vector $\mathbf{v}$ in the origin of a coordinate system of a three-dimensional (3-D) Euclidean space. The coordinates depend on the choice of the **versors** or **unit-vectors**² describing the points in space (see Fig. 2.1):

$$\mathbf{v} = v_1 \mathbf{i} + v_2 \mathbf{j} + v_3 \mathbf{k}. \quad (2.3)$$

A different choice of the versors corresponds to changing the basis of the Euclidean space and, then, of $\mathbb{R}^3$. The new vector $\mathbf{v}'$ expressed in the new basis is:

$$\mathbf{v}' = v'_1 \mathbf{i}' + v'_2 \mathbf{j}' + v'_3 \mathbf{k}'. \quad (2.4)$$

The real numbers v_1, v_2, v_3 are called **coordinates**. In classical mechanics, coordinates are often labeled as v_x, v_y, v_z . The basis is described by three perpendicular versors such that $\mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{i} = 0$. In linear algebra, this is equivalent to changing the “orthonormal basis” $\mathbf{i}, \mathbf{j}$, and $\mathbf{k}$ with a new “orthonormal basis” $\mathbf{i}', \mathbf{j}'$, and $\mathbf{k}'$ (Larson and Falvo, 2009).

In a n -dimensional space, the coordinates of $\mathbf{v}$ are an n -tuple of real numbers $v_1 \cdots v_n$ associated to the basis $\mathbf{e}_1 \cdots \mathbf{e}_n$. We can write $\mathbf{v}$ in a neater form as:

$$\mathbf{v} = \sum_{i=1}^n v^i \mathbf{e}_i. \quad (2.5)$$

Even better, we can employ the

Einstein summation convention: when an index variable appears twice in a single term, it implies summation of that term over all the values of the index.

The vector is then:

$$\mathbf{v} = v^i \mathbf{e}_i \quad (2.6)$$

which is a shortening of eqn 2.5. Ignore for the moment that the index of the coordinates is a superscript (v^i) and not a subscript (v_i): it will be clearer in a minute. A change of basis in $\mathbb{R}^n$ is a linear transformation that maps each versor $\mathbf{e}'_i$ of the new basis to a linear combination of the

²A versor in physics (not to be confused with versors in mathematics) is a unit-norm vector that has the same direction of the vector. A versor is labeled in bold letters with a hat: for instance, $\hat{\mathbf{r}}$. In this book, the versors of the reference-frame axes are indicated without the hat as $\mathbf{i}$ (x -axis), $\mathbf{j}$ (y -axis), and $\mathbf{k}$ (z -axis); see Fig. 2.1. For a general n dimensional space, the axes are n and the corresponding versors are labeled $\mathbf{e}_1, \dots, \mathbf{e}_n$.

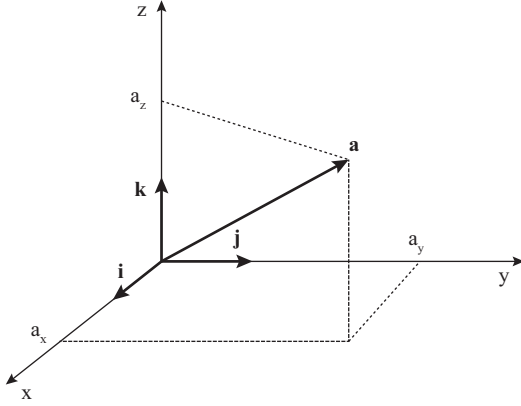


Fig. 2.1 A vector $\mathbf{a}$ starting from the origin of the reference frame. The coordinates of the vector are the projections of $\mathbf{a}$ in three arbitrary and perpendicular axes: x , y , and z . These projections are called $a_x \equiv a_1$, $a_y \equiv a_2$, and $a_z \equiv a_3$. The unit vectors (versors) parallel to the axes are $\mathbf{e}_1 \equiv \mathbf{i}$, $\mathbf{e}_2 \equiv \mathbf{j}$, and $\mathbf{e}_3 \equiv \mathbf{k}$, respectively.

old versors $\mathbf{e}_i$:

$$\mathbf{e}'_i = \sum_{j=1}^n b^j_i \mathbf{e}_j \equiv b^j_i \mathbf{e}_j \quad (2.7)$$

We can also describe a change of basis using the matrix formalism. Equation 2.7 reads

$$\mathbf{e}' = B\mathbf{e} = b^j_i \mathbf{e}_j \quad (2.8)$$

where B is a $n \times n$ matrix, b^j_i are the coefficients of the matrix – that is, the number in row j and column i – and $\mathbf{e}$ is a column of n versors.

If the vector $\mathbf{v}$ must be independent of the choice of the base, we cannot leave the coordinates unchanged, otherwise the position of the vector in space would change too. The coordinates must change to compensate for the change of the versors or, equivalently,

the components of the vector that corresponds to the coordinates must *contra-vary* to compensate for the changes in the basis and keep the vector fixed in space.

Since the change of basis makes:

$$\mathbf{v} = v^i \mathbf{e}_i = \mathbf{v}' = v'^i \mathbf{e}'_i = v'^i b^j_i \mathbf{e}_j, \quad (2.9)$$

the coordinates must change as:

$$v'^i = a^i_j v^j \quad (2.10)$$

to have a vector fixed in space. In eqn 2.10, a^i_j are the $n \times n$ elements of the matrix A , which is the *inverse* of B ($A = B^{-1}$). In summary:

- after a change of basis described by the matrix B , the coordinates of a vector transform as $\mathbf{v}' = A\mathbf{v}$, where $A = B^{-1}$. All n -tuples that transform in this way are called **contravariant vectors** or simply **vectors**, and are written with upper indexes (v^i).
- all n -tuples that transform as the versors – which means as one particular choice of the basis for a given space – after a change

of basis ($v'_i = Bv_i$) are called **covariant vectors** and are written with lower indexes (v_i).

So covariance and contravariance describe how certain geometric or physical objects change after a change of basis. We require a vector to be base-independent to get a good candidate for a real physical quantity. The constraint of base-independence implies the covariance of the vectors and the contravariance of the coordinates. These are non-trivial constraints: a vector is no more a simple n -tuple of real numbers but must transform in a well-defined way if we decide to change the reference frame.

2.1.2 Classification of covariant objects

We can further generalize this approach classifying all base-independent objects by their behavior after a change of basis.

Scalars are quantities that do not change after a change of basis
Contravariant vectors, or simply **vectors**, are n -tuples that transform as $x^{i'} = a^{i'}_i x^i$.
Contravariant tensors of rank 2 are a $n \times n$ set of numbers that transform as $T^{i'j'} = a^{i'}_i a^{j'}_j T^{ij}$.
Covariant vectors are n -tuples that transform as $x_{i'} = b^i_{i'} x_i$.
Covariant tensors of rank 2 are a $n \times n$ set of numbers that transform as $T_{i'j'} = b^i_{i'} b^j_{j'} T_{ij}$.

A rank-2 tensor looks like a $n \times n$ matrix but the n^2 coefficients are carefully chosen to fulfill the definitions above. Further generalizations (e.g. tensors of rank- n , spinors, etc.) are also available but are not needed at this stage. Just out of pedantry, note that the definition of a rank- n tensor is straightforward: a contravariant tensor of rank n is a set of numbers that transforms as:

$$T^{i'_1 i'_2 \dots i'_n} = a^{i'_1}_{i_1} a^{i'_2}_{i_2} \dots a^{i'_n}_{i_n} T^{i_1 i_2 \dots i_n} \quad (2.11)$$

and we can also build mixed covariant–contravariant tensors like:

$$T^{i'_1 i'_2 \dots i'_n}_{j'_1 j'_2 \dots j'_m} = a^{i'_1}_{i_1} a^{i'_2}_{i_2} \dots a^{i'_n}_{i_n} b^{j_1}_{j'_1} b^{j_2}_{j'_2} \dots b^{j_m}_{j'_m} T^{i_1 i_2 \dots i_n}_{j_1 j_2 \dots j_m}. \quad (2.12)$$

Physicists are particularly interested in changes of basis that represent smooth (i.e. differentiable) coordinate transformations. The Lorentz transformations in SR and the Galilean transformations in classical physics (see Sec. A.1) are two notable examples. For a generic smooth coordinate transformation between $\mathbf{x}'$ and $\mathbf{x}$, we have:

$$a^{i'}_i = \frac{\partial x^{i'}}{\partial x^i} \quad (2.13)$$

and

$$b^i_{i'} = \frac{\partial x^i}{\partial x^{i'}} \quad (2.14)$$

so a (contravariant) vector transforms as:

$$v^{i'} = \frac{\partial x^{i'}}{\partial x^i} v^i. \quad (2.15)$$

You may wonder whether there are interesting physical objects that transform as covariant vectors or contravariant tensors. There are many. Quantities that involve *odd derivatives* of vectors are often covariant. For instance, the **gradient** of a generic function $f(x^1, \dots, x^n)$ of the coordinates is a covariant vector. This statement follows from:

$$\frac{\partial f}{\partial x^{i'}} = \frac{\partial f}{\partial x^i} \frac{\partial x^i}{\partial x^{i'}} = \frac{\partial f}{\partial x^i} b^i_{i'}. \quad (2.16)$$

In classical mechanics, the inertia tensor is a rank-2 contravariant tensor that transforms as a tensor for rotations in space (Goldstein, 1980). Relativistic electrodynamics is also built upon covariant and contravariant rank-2 tensors under Lorentz transformations, as discussed in Chap. 6.

2.2 Particles in space-time

General covariance can be applied to the specific case of SR. Here, the space is four-dimensional (time and the three directions in space) and the smooth changes of basis correspond to the Lorentz transformations. For a reference system O' moving with a velocity $v\mathbf{i}$ in the x direction relative to the system O (Fig. 2.2), the Lorentz transformations in SI read:

$$\begin{aligned} t' &= \gamma(t - \frac{\beta}{c}x) \\ x' &= \gamma(x - \beta ct) \\ y' &= y \\ z' &= z \end{aligned} \quad (2.17)$$

where β is equal to v/c and called the **β Lorentz factor**.³ In NU, $\beta = v$.

$$\gamma \equiv \frac{1}{\sqrt{1 - \beta^2}} \quad (2.18)$$

³Equation 2.17 is just an example of Lorentz transformations and is called a **Lorentz boost along $\mathbf{x}$** . The general Lorentz transformations are described and classified in Sec. A.1.

is called the **γ Lorentz factor** and is $\sqrt{1/(1 - v^2)}$ in NU.

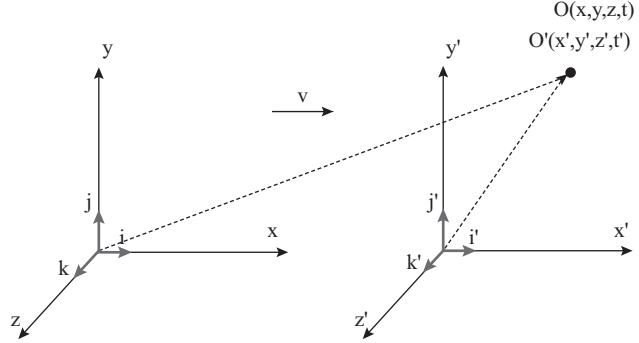
SR mixes space with time but only in the direction of motion of the system O' relative to O . The coordinates in the plane perpendicular to this direction are the same in both systems. To embed the covariant formalism in eqn 2.17, we define the coordinates of (contravariant) vectors in $\mathbb{R}^4$ starting from 0 instead of 1:

$$x^\mu = (x^0 \equiv ct, x^1 \equiv x, x^2 \equiv y, x^3 \equiv z) \quad (2.19)$$

x^μ is called a **four-vector**. In NU,

$$x^\mu = (x^0, x^1, x^2, x^3) \quad (2.20)$$

Fig. 2.2 A reference system O' moves in the rightmost direction along the x axis ($\mathbf{v} = v\mathbf{i}$) with respect to a system O . The point on the right shows an event and its coordinates in O and O' .



and

$$\begin{pmatrix} x'^0 \\ x'^1 \\ x'^2 \\ x'^3 \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}. \quad (2.21)$$

Using the Einstein summation convention, we can write eqn 2.21 as:

$$x'^\mu = \Lambda^\mu_\nu x^\nu \quad (2.22)$$

where Λ^μ_ν is the number located in row μ and column ν of the 4×4 matrix of eqn 2.21. As a consequence, the matrix A that represents the change of coordinates in $\mathbb{R}^4$ has a very specific form given by Λ^μ_ν . In the language of general covariance, x^μ is a contravariant vector for the change of coordinates represented by the Lorentz transformations Λ^μ_ν . We can get the inverse Lorentz transformation from O to O' using eqn 2.17 by changing the sign of the velocity:

$$\begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma & 0 & 0 \\ \beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x'^0 \\ x'^1 \\ x'^2 \\ x'^3 \end{pmatrix}. \quad (2.23)$$

As expected, the matrix associated with eqn 2.23 is Λ^{-1} .

If the direction of motion of the two systems is not along the x -axis but on a generic unit-vector $\mathbf{v}/v$, we can define $\boldsymbol{\beta} = \mathbf{v}/c$ (or $\boldsymbol{\beta} = \mathbf{v}$ in NU). The corresponding Lorentz transformations, called **Lorentz boosts**, are now:

$$\begin{aligned} x'^0 &= \gamma(x^0 - \frac{\mathbf{v} \cdot \mathbf{x}}{c}) \\ \mathbf{x}' &= \mathbf{x}_\perp + \gamma(\mathbf{x}_\parallel - \frac{\mathbf{v}x^0}{c}) \end{aligned} \quad (2.24)$$

in SI. Here, $\mathbf{x}'_\perp$ and $\mathbf{x}'_\parallel$ are the spatial components projected in the direction perpendicular and parallel to the motion. We can write eqn 2.17 in NU in a compact form employing the matrix notation of linear algebra:

$$\begin{pmatrix} x'^0 \\ \mathbf{x}' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\boldsymbol{\beta}^T \\ -\gamma\boldsymbol{\beta} & \mathbb{I} + (\gamma - 1)\frac{\boldsymbol{\beta}\boldsymbol{\beta}^T}{\beta^2} \end{pmatrix} \begin{pmatrix} x^0 \\ \mathbf{x} \end{pmatrix} \quad (2.25)$$

where $\mathbb{I}$ is the 3×3 identity matrix,

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_1 = v_x \\ \beta_2 = v_y \\ \beta_3 = v_z \end{pmatrix}$$

is a column-vector and

$$\boldsymbol{\beta}^T = (\beta_1, \beta_2, \beta_3) \quad (2.26)$$

is a row-vector. These compact notations may be puzzling for a novice. In the covariant formalism of SR, we should just use covariant quantities like the x^μ vectors. However, it is useful to distinguish the space components of x^μ from the time component because space can be measured with rulers, time with clocks, and real experiments do not provide straightforward space-time measurements. In this sort of mixed notation, the space components are written with *Latin indexes*: $\mathbf{x} = (x^1, x^2, x^3)$. The position of a particle at a time t is labeled $\mathbf{x}$ as in classical physics and corresponds to the SR “event” $x^\mu = (ct, \mathbf{x})$. The **event** $x^\mu = (ct, \mathbf{x})$ is the actual measurement in space-time and it tells us where and when we find the particle: in the position $\mathbf{x}$ at the time t . Events are always labeled with *Greek indexes* ($\mu = 0, 1, 2, 3$) to avoid any source of confusion with space coordinates.

2.2.1 The Minkowski space-time

The covariant formalism makes very clear the rich mathematical structure of SR. It also provides simple means to construct relativistic dynamics in analogy to classical physics, as shown in Sec. A.3.

The first observable we can build using covariance is the **trajectory** of a point-like particle. In classical physics, the trajectory is a function that maps the position in space as a function of time. Since time is universal, the distance between two points is the same for any observer even if the coordinates of the points change among observers. The (squared) distance traveled between t and $t + dt$ is thus the same in any inertial frame and is equal to:

$$ds^2 = dx^2 + dy^2 + dz^2 \quad (2.27)$$

where

$$d\mathbf{x} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}. \quad (2.28)$$

In SR, the observers disagree on time intervals and on the distance between two points in space.⁴ However, they agree on the squared difference between time and space. If an observer in O releases a particle at time t_1 in $\mathbf{x}_1$, records the passage of the particle in a detector located in $\mathbf{x}_2$ at time t_2 , and compares her findings with an observer O' , SR ensures that:

$$\begin{aligned} & c^2(t_2 - t_1)^2 - (x_2 - x_1)^2 - (y_2 - y_1)^2 - (z_2 - z_1)^2 \\ &= c^2(t'_2 - t'_1)^2 - (x'_2 - x'_1)^2 - (y'_2 - y'_1)^2 - (z'_2 - z'_1)^2 \end{aligned}$$

⁴This is a consequence of time dilation and space contraction, as shown in sec. A.2.

(see Exercise 2.3). We can then replace eqn 2.27 with

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 \quad (2.29)$$

and be certain that every observer measuring ds^2 gets the same number, whatever frame she trusts. In the language of covariance, ds^2 is a Lorentz scalar, that is it is **invariant under Lorentz transformations**. Similarly, the distance between two events is a scalar: $c^2(t_2 - t_1)^2 - (x_2 - x_1)^2 - (y_2 - y_1)^2 - (z_2 - z_1)^2$. If $t_1 = 0$ and $\mathbf{x}$ is the origin of the system, $c^2 t^2 - x^2 - y^2 - z^2 = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2$ is a scalar too.

The quantity $(x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2$ can be expressed for any contravariant vector as:

$$g_{\mu\nu} x^\mu x^\nu \quad (2.30)$$

employing the Einstein summation convention. The 4×4 real matrix $g_{\mu\nu}$ is the **metric tensor** of SR:

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (2.31)$$

The metric tensor of eqn 2.31 distills the essence of the covariant formalism *for the particular case of SR*. $g_{\mu\nu}$ fulfils three important properties:

- $g_{\mu\nu}$ is constant. Unlike, for example, GR (Cheng, 2010), eqn 2.31 holds in any point of the space and at any time, so that $g_{\mu\nu}(x^\mu) = g_{\mu\nu}$.
- $g_{\mu\nu}$ is a diagonal matrix and differs from the identity matrix only by a sign in the space $\mu = 1, 2, 3$ components.
- $g_{\mu\nu}$ is equal to its inverse.

The peculiar form of the metric tensor provides a rule to transform a contravariant quantity into a covariant quantity:

for a generic contravariant four-vector a^μ ,

$$a_\mu = g_{\mu\nu} a^\nu \quad (2.32)$$

is covariant. As a consequence, the SR rule that transforms a contravariant vector in a covariant vector is:

$$\begin{aligned} a_0 &= a^0 \\ a_i &= -a^i \end{aligned} \quad (2.33)$$

for $i = 1, 2, 3$.

Of course, such a simple rule holds in SR, only. We can use the same rule to transform a contravariant tensor in a covariant tensor:

$$T_{\mu\nu} = g_{\mu\alpha} g_{\nu\beta} T^{\alpha\beta} \quad (2.34)$$